

Effects of Soil Moisture Variability on Maize Yield: A Field Study

Abstract: This study examines the relationship between soil moisture variability and maize (*Zea mays*) yield across 48 experimental plots in semi-arid farmland over two growing seasons (2024-2025). Soil moisture was monitored at 15-centimeter depth using capacitance probes, and yield was measured at harvest for each plot.

Results indicate that plots maintaining soil moisture between 22% and 30% volumetric water content during the critical tasseling stage produced significantly higher yields, averaging 7.8 tonnes per hectare, compared to 5.1 tonnes per hectare in plots that experienced moisture stress (below 15%) during the same growth stage.

Effects of Soil Moisture Variability on Maize Yield: A Field Study

The study also found that irrigation timing mattered more than total irrigation volume. Plots receiving the same total water amount but distributed in smaller, more frequent intervals during tasseling outperformed plots receiving infrequent, heavy irrigation by an average of 11%.

These findings suggest that for smallholder farmers in semi-arid regions, optimizing irrigation scheduling around the tasseling growth stage may be more cost-effective than simply increasing total water usage. Future research should examine whether these patterns hold across different maize hybrids and soil types.